

Research papers

Energy management approach for wayside energy storage system in urban rail transit considering real-observable characteristics: A deep reinforcement learning method based on fuzzy logic guided

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ABSTRACT

The deployment of wayside energy storage system (ESS) in urban rail transit (URT) facilitates the efficient utilization of regenerative braking energy of trains, making it a widely adopted solution in the realm of energy-saving technology for urban rail transportation. Given the pronounced nonlinear and time-varying characteristics of traction power supply system (TPSS), deep reinforcement learning (DRL) demonstrates significant potential for optimizing ESS control and energy management. However, due to the incomplete observability of the environment in practical TPSS applications, conventional DRL methods encounter challenges in reward design, environmental exploration, and strategy learning. To address these limitations, this paper proposes a DRL approach guided by fuzzy logic, designed to improve the agent's learning efficiency in identifying optimal strategies. Compared to previous DRL methods applied to ESS energy management in urban rail systems, the fuzzy logic guided agent demonstrates a substantial enhancement in energy-saving performance.

1. Introduction

1.1. Background and motivation

The power consumption of urban rail transit (URT) remains a significant contributor to overall energy usage. With the widespread adoption of regenerative braking technology in URT trains [1], maximizing the utilization of regenerative braking energy has become a key focus in energy conservation for urban rail systems. Statistics show that regenerative braking energy can constitute 20–40 % of traction energy consumption [2,3]. Wayside energy storage system (ESS) is designed to absorb and store the regenerative braking energy of the traction power supply system (TPSS) and release it when required during traction, enabling efficient energy utilization [4–6]. This not only reduces operational energy consumption and operating costs but also decreases carbon emissions, supporting the sustainable and efficient development of urban rail transit. A typical wayside ESS in URT typically consists of energy storage components and energy storage converters (commonly bidirectional DC-DC converters) [7]. The energy management strategy (EMS), serving as the top-level control strategy for ESS, plays a pivotal role in determining its operational state, including the activation and

deactivation of energy storage converter, output power control, and the transition between charging and discharging modes. The primary objective of EMS is to capture residual regenerative braking energy and release it during traction to achieve energy-saving goals [4,8–10]. As a result, optimizing ESS control and energy management optimization is crucial for maximizing energy savings.

However, due to the rapid movement of multiple trains causes constant changes in the system topology, resulting in the urban rail TPSS exhibiting strongly nonlinear and time-varying characteristics. Additionally, under practical engineering conditions for the wayside ESS control, the TPSS environment is often only partially observable. These two factors significantly impede further research and practical applications of energy management for urban rail ESS.

1.2. Related work

Currently, energy management strategies (EMS) for utilizing regenerative braking energy in urban rail ESS are widely studied and can be broadly divided into three categories. Considering the incomplete observability of TPSS, the first category involves using traction network voltage or other indirect variables as references to formulate simple

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fixed rules for ESS energy management. This approach is commonly implemented in practical scenarios [11–14]. For instance, authors in [14] establish charging and discharging voltage thresholds for ESS based on the substation's no-load voltage and determines ESS operating conditions by comparing these thresholds with real-time traction network voltage. Historically, it was believed that variations in traction network voltage reflected train operating conditions: energy consumption during traction causes voltage drops, while residual regenerative braking energy during braking leads to voltage rises. However, in real-world scenarios with multiple substations within a traction power supply section, factors such as fluctuations in no-load voltage at individual substations and differences in no-load voltage across multiple substations make it challenging to accurately infer train operating conditions across the entire section based solely on voltage observations from a single station [15]. In an idealized scenario, the second category of EMS directly evaluates regenerative braking energy in the TPSS by acquiring real-time train status and subsequently controlling the ESS. In [16], a method is proposed to adjust ESS control by observing the activation characteristics of onboard braking resistors in real time. Similarly, [17] optimizes ESS control strategies by tracking the real-time traction and braking power of train motors. In [18], a collaborative model for the ESS and train operation system is established using a discrete-time measurement method, enabling joint optimization of ESS control and train timetables. Additionally, [19] proposes equating the aggregate power of all trains within a power supply section to the residual power after power interactions, using this as a basis to guide the charging and discharging operations of ESS. However, practical implementation faces significant challenges due to the stringent requirements of train-to-ground wireless communication technologies in terms of bandwidth, reliability, and latency. The presence of multiple trains within a single power supply section further complicates the integration of communication-based train control (CBTC) into ESS, particularly given concerns related to information security and technical retrofitting. Furthermore, the energy-saving performance of these two EMS categories remains relatively limited due to the lack of effective optimization processes.

In pursuit of enhanced energy-saving efficiency, researchers have increasingly explored energy management strategies that integrate various optimization algorithms. These include heuristic algorithms, such as genetic algorithms (GA), and model-based approaches, such as dynamic programming (DP). The objective is to identify suboptimal or optimal operating points by optimizing ESS control parameters through these algorithms. For instance, literature [20] proposed an EMS using mixed-integer second-order cone programming (MISOCP) to optimize the charging and discharging power of wayside ESS (WESS), achieving improved economic performance. In literature [21], the Pontryagin minimum principle was employed to determine the global optimal power distribution for hybrid energy storage systems (HESS). Authors in [22] applied a multi-objective (MO) optimization approach to enhance ESS energy management for regenerative braking energy recovery. Similarly, reference [23] utilized fuzzy logic control (FLC) rules to manage ESS operations based on the energy variation between the substation and the ESS. A comprehensive review in [24] analyzed the application of GA and its variants in electrified railway and urban rail transit systems, particularly in ESS energy management. This review also discussed challenges faced by GA and hybrid algorithms combining particle swarm optimization (PSO), ant colony optimization (ACO), and FLC. Despite their promising potential, these optimization-based methods are hindered by several challenges. The time-varying characteristics of TPSS, the uncertainty of train load conditions, and difficulties in accurately acquiring system parameters often lead to issues such as challenges in model establishment, parameter inaccuracies, complex optimization problem resolution, and limited generalizability in online applications.

Given the strong nonlinear and time-varying characteristics of urban rail TPSS, deep reinforcement learning (DRL), with its advanced

capabilities in environmental perception and continuous strategy optimization, is highly suitable for addressing challenges in ESS control and energy management [25]. In recent years, several studies have explored the application of DRL in this domain [26–31]. For instance, literature [26] proposed an online EMS optimization method based on the deep Q-learning (DQN) algorithm. The change in energy-saving rates before and after ESS deployment was utilized as the reward function to guide the agent's learning process. Additionally, the train power and position within the TPSS were incorporated as input states. However, ensuring consistency between TPSS states before and after ESS installation poses significant challenges, raising concerns about the validity of current energy-saving evaluation methods, which will be elaborated upon in the Problem Formulation Section. Moreover, due to the incomplete observability of the TPSS in practical engineering conditions, using real-time train power and position as agent input states is highly impractical. Literature [27] employed the deep deterministic policy gradient (DDPG) algorithm, which offers enhanced performance. However, during online optimization, train power, position, and other variables—often difficult to observe in practice—were assumed to be readily available. Prediction tasks based on these assumptions were further used to inform the agent's reward and state design, creating a gap between theoretical and practical applicability. In [28], an annealing bias-priority experience replay twin delayed deep deterministic policy gradient algorithm (A-TD3) was proposed for optimizing the power allocation strategy of onboard HESS. Inputs such as train acceleration and traction motor speed were used for state selection, which is theoretically feasible for onboard ESS due to local sampling capabilities. However, applying this approach to wayside ESS remains challenging. Additionally, similar to [26], the reward function was based on the TPSS energy-saving rate, inheriting the same limitations concerning inconsistent state evaluation.

Based on the above analysis, it is evident that the inherent characteristics of TPSS significantly limit the optimization effectiveness of rule-based, heuristic, and model-based methods for urban rail ESS energy management. Furthermore, these approaches face considerable challenges in practical engineering applications. At least for now, deep reinforcement learning (DRL) represents the most advanced research direction and promising application in this field. However, the incomplete observability and instability of the TPSS environment, coupled with the frequent short-distance starts and stops of urban rail trains, result in rapid generation and consumption of regenerative braking energy. These factors introduce significant challenges to the deployment and training of DRL-based energy management strategies. Key challenges include agent exploration, reward shaping, and action execution. This study identifies the following critical gaps that need to be addressed:

- 1) **Incomplete observability and instability:** How can DRL methods be improved and adapted for wayside ESS energy management under the incomplete observability and instability inherent in real-world TPSS environments?
- 2) **Efficient learning in real TPSS environments:** How can agents be guided to learn efficiently within practical conditions, enabling to make accurate and optimal decisions?
- 3) **Algorithm scalability and adaptability:** How can the proposed method be made compatible with various DRL algorithms (value-based and policy-based) while ensuring scalability to URT with diverse operational characteristics?

1.3. Contributions

In this paper, we analyze the energy distribution within urban rail TPSS. Based on the observable energy distribution relationships in practice, we perform inferential analysis on the regenerative braking demand and traction demand in the TPSS. By combining expert experience with fuzzy logic reasoning, we propose an agent learning method guided by fuzzy logic. This method, which focuses on the energy-saving

effect of URT ESS energy management, effectively addresses the need for efficient utilization of regenerative braking energy in URT, thereby enhancing the learning efficiency of the DRL agent in optimizing the energy management strategy. The specific contributions of this paper are as follows:

- 1) **Integration of fuzzy logic with reward shaping:** For the first time, fuzzy logic is integrated with agent reward shaping. The qualitative relationship between energy changes in TPSS and the regenerative braking energy recovery effect is identified. This relationship is then translated into specific rules based on fuzzy logic, enabling the design of rewards to guide the agent's learning process.
- 2) **Development of a novel DRL framework for ESS energy management:** A DRL framework suitable for optimizing the control of energy management in urban rail ESS is established. The proposed agent learning method shows improved performance when applied to two classical DRL algorithms, DQN and DDPG.
- 3) **Enhanced adaptability to partially observable TPSS:** Compared with traditional DRL, the proposed method improves adaptability in real-world TPSS with limited observability. This enables more effective learning of energy management strategy, leading to superior regenerative braking energy utilization and reduced braking resistor energy loss.

In this study, several key assumptions are made to simplify the complexity of the URT system and focus on the energy management of the ESS. First, the TPSS environment is assumed to be partially observable, meaning only certain variables (e.g., substation and ESS state) are directly measurable. Additionally, the ESS is assumed to operate independently. Finally, the fuzzy logic rules for reward shaping are based on expert experience and qualitative analysis of energy flow in the TPSS. Detailed parameter conditions for the case study, including line, train, ESS, and agent parameters, are provided in the Appendix (Tables A1–A5). These assumptions ensure the tractability of the problem while maintaining relevance to real-world URT systems.

1.4. Paper organization

Section 2 discusses the limitations of current DRL methods in optimizing the control of ESS for regenerative braking energy absorption. Section 3 provides a detailed explanation of the design principles and process of the proposed fuzzy logic guided agent learning method. Section 4 presents practical case studies and verifies the effectiveness of the method. Finally, Section 5 summarizes the paper.

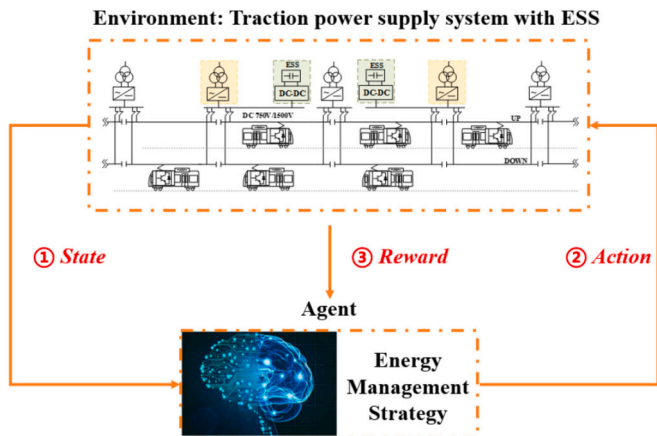


Fig. 1. DRL framework for energy management of URT ESS.

2. Problem formulation

The basic DRL framework applied to energy management of urban rail ESS is shown in Fig. 1. Reward shaping is one of the most important issues in DRL, as it directly influences the exploration, convergence accuracy, training speed, and determines whether the agent can learn effective strategies.

In practical scenarios, when applying DRL to optimize ESS energy management, rewards shaping becomes a challenge. Since the energy-saving effect is the most crucial and commonly pursued objective, the substation energy-saving rate within the agent exploration step Δe has been common to use as reward in past work. The reward and the current prevalent definition of substation energy saving rate e are as follows

$$\begin{cases} e = \left(1 - \frac{E_{sub_ess}}{E_{sub_noess}}\right) \times 100\% \\ r = \Delta e \end{cases} \quad (1)$$

where E_{sub_noess} and E_{sub_ess} represent the total output energy of the substations before and after configuring ESS, respectively. r is for agent reward.

In this work, takes the power supply section of a subway line in Beijing, China, which includes five substations as an example to analyze the energy distribution of TPSS. Line conditions are shown in Table A1 in the Appendix, with a simulated line of 6290 m. Train and ESS operation parameters can be found in Tables A2 and A3 in the Appendix. The operation process of the trains on the line takes the departure interval as a cycle. Taking a 7-min cycle as an example, the operation curve of one train is shown in Fig. 2.

Fig. 3 shows the energy distribution of substations and braking resistors before and after the ESS configuration during a 7-minute departure interval. In figure, ΔE_{sub} represents the substations energy consumption for each time period, and each time period is 30 s. ΔE_{bR} is the energy loss of braking resistors. In the 1th, 2th, 14th time periods, ΔE_{sub_noess} is obviously less than ΔE_{sub_ess} , while ΔE_{bR_noess} is greater than or equal to ΔE_{bR_ess} . In this case, the calculated energy-saving rate of the substations is negative. However, in reality, the ESS absorbs regenerative braking energy and reduces the energy loss of braking resistors, which is essentially energy-saving. Therefore, in certain specific situations, relying on substation energy-saving rate may not accurately reflect the energy-saving effect. And based on this as the goal for ESS control will inevitably impact the effectiveness of online optimization.

From another perspective, it's essential to note that the E_{sub_noess} in Eq. (1) is not fixed. The environmental state of TPSS before and after the ESS configured cannot be guaranteed to be completely consistent, which poses significant challenges to determining the value of E_{sub_noess} . Fig. 4 (a) shows the substations energy distribution when the departure interval is the same (that is, the train operation conditions keep unchanged) but the no-load voltage characteristics of the substations are different. It can be seen that even under the same load conditions, the

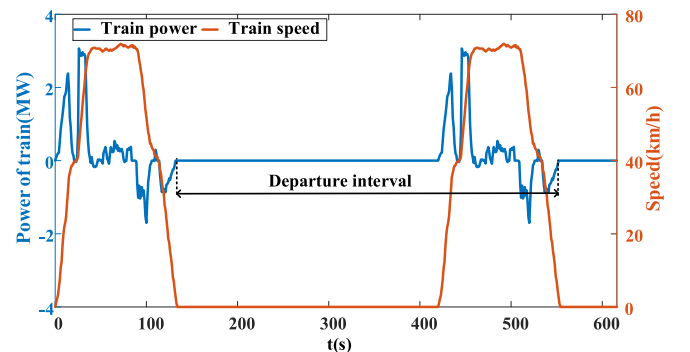
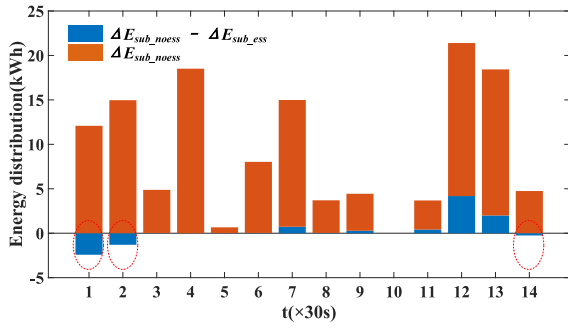
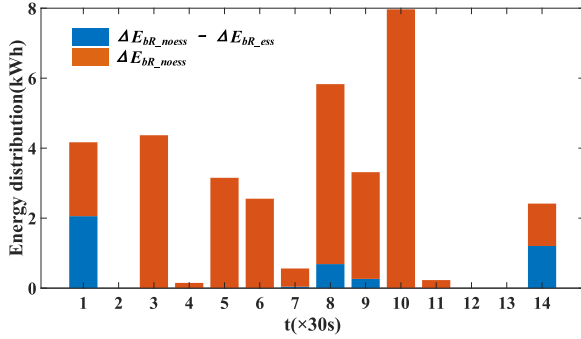


Fig. 2. The operation curve of the train on up line.

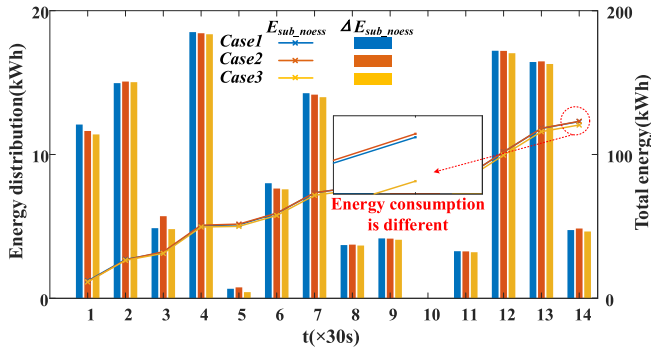


(a) Energy consumption distribution of substations.

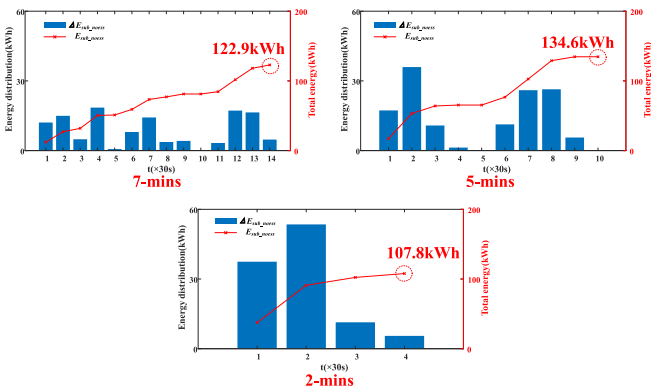


(b) Energy consumption distribution of braking resistors.

Fig. 3. Energy distribution in TPSS with and without the ESS.



(a) Under different no-load voltage characteristic.



(b) Under different departure interval.

Fig. 4. Energy distribution of substations without ESS.

distribution of $E_{sub, noess}$ will change due to the instability of the substation's own characteristics. Fig. 4 (b) shows the energy distribution of the substation with different departure intervals (that is, the load characteristics change) but the no-load voltage characteristics are the same. In the actual condition, factors such as frequent adjustments of departure intervals, changes in passenger flow and the driver manual operation will lead to the real-time change of train load characteristics. Therefore, how to assess energy-saving effect online is one of the pressing research issues that needs to be addressed. When the energy management of ESS is optimized by DRL, designing rewards based on Eq. (1) or its variants will make it difficult for the agent to achieve the desired effect in practical application and training.

At the same time, the environmental state of URT TPSS is not completely observable in engineering conditions. In [19], the authors established the agent state as follow, including the positions and powers of all trains near the ESSs.

$$s = [d_1, p_1, \dots, d_N, p_N, SOC, \dots] \quad (2)$$

where d_k and p_k represent the displacement and power of the k-th train, respectively.

However, for real-time control of wayside ESS, these train states are difficult to obtain. As mentioned in the previous section, observables typically include only ground information. Therefore, even if agent adopts the same actions, the size and even the positive-negative relationship of reward may differ in the same states. In summary, the current methods based on substation energy consumption to design agent reward functions have significant flaws and limitations. At the same time, due to the partially observable of TPSS, it is challenging to develop reward based on factors such as energy loss of onboard braking resistors, regenerative braking energy distribution and so on. Therefore, in practical applications, agents face the dilemma of low efficiency and difficulty in learning during the process of optimizing ESS energy management strategy.

3. An agent learning method based on fuzzy logic guided

This paper proposed an agent learning method based on fuzzy logic guided. It formulates fuzzy logic rules entirely based on the energy relationships between observables in practical scenarios to guide the learning process of the agent.

3.1. Modeling of urban rail transit TPSS

The TPSS mainly includes three energy interaction subjects, 1) substation, 2) train, and 3) ESS.

- 1) **Substation:** Two 12-pulse rectifier units in the traction substation are connected in parallel to form 24-pulse rectifier to supply power to the TPSS. Considering the output characteristic curve of the rectifier, the equivalent model of substation can be replaced by an ideal voltage source and equivalent internal resistance in series [32]. The equivalent model of substation is shown in the Fig. 5 and its volt-ampere characteristic equation is shown in Eq. (3).

$$u_{sub} = u_0 - r_s \cdot i_{sub}, i_{sub} \geq 0 \quad (3)$$

where u_0 and r_s are the no-load voltage and equivalent resistance of the substation.

- 2) **Train:** When the train operates in traction condition, it absorbs energy from traction network. When the train is braking, the dynamic energy is converted into electric energy based on regenerative braking technology and released into the traction network [5]. It is important to note that the pantograph voltage will rise rapidly when the train brakes and there is no adjacent traction train or ESS to absorb braking energy. In order to prevent overvoltage, the braking

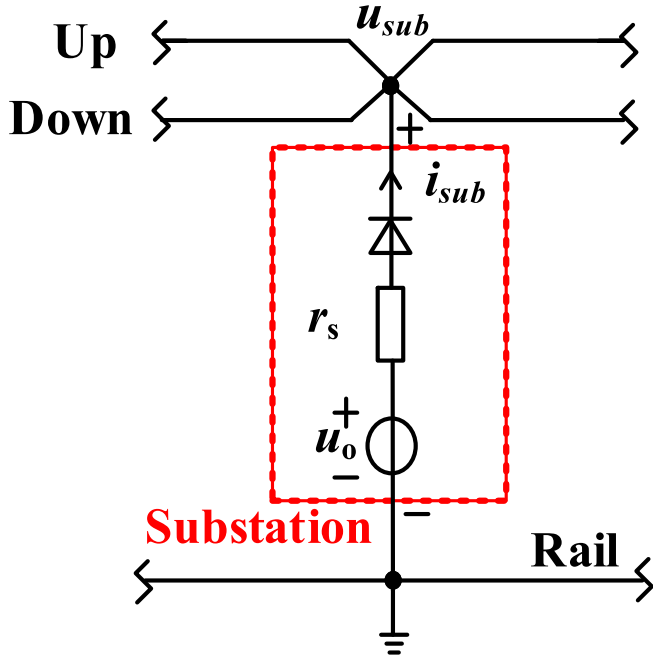


Fig. 5. The equivalent model of substation.

resistor will start and consume part of the braking energy. Therefore, the train is modeled as a power source with the on-board braking resistor, as shown in Fig. 6. The braking resistor adjusts the on/off of the brake chopper according to the voltage of the filter capacitor on the DC side of the traction converter, and changes its equivalent resistance.

- 3) ESS: This paper discusses the wayside ESS based on supercapacitor. Based on the first-order circuit model, the supercapacitor is modeled as a series voltage source with the equivalent internal resistance [10], as seen in Fig. 7. The SOC expression of the supercapacitor and the terminal voltage range in normal operation are shown in Eq. (4).

$$\begin{cases} u_{sc,min} < u_{sc} < u_{sc,max} \\ SOC = u_{sc}^2 / u_{sc,max}^2 \end{cases} \quad (4)$$

where $u_{sc,min}$ and $u_{sc,max}$ are the limit values of the SC terminal voltage. And SC stops working when the voltage of SC is less than $u_{sc,min}$ or greater than $u_{sc,max}$.

3.2. Energy relationship between observables in TPSS

When applying DRL to the online optimization for energy management of URT ESS, the practical problem to be considered is how to effectively utilize observables and use them for reward shaping. The observable energy flow paths in the TPSS are limited to between the substations and ESSs. By ignoring smaller energy loss components

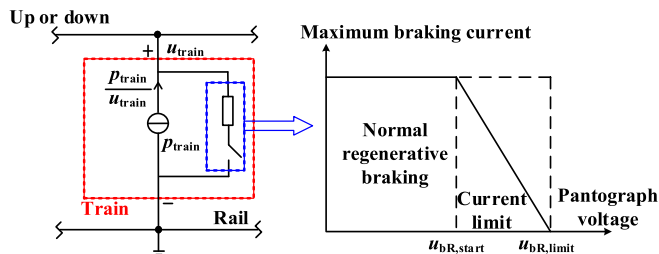


Fig. 6. The equivalent model of train.

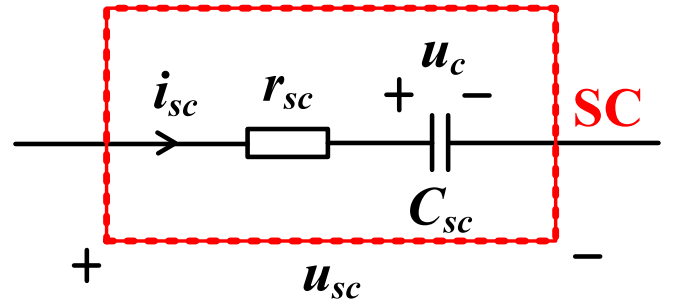


Fig. 7. The equivalent model of supercapacitor.

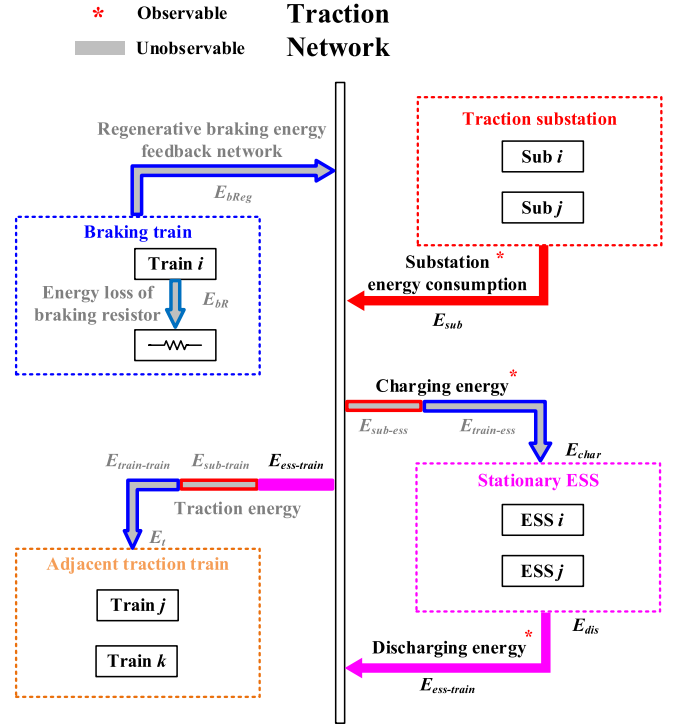


Fig. 8. Diagram of energy flow of TPSS with ESS.

within the system such as the lines, rectifier units, and so on [4]. The energy flow in the system can be illustrated as shown in Fig. 8.

The formula for energy flow paths of observables is given by Eq. (5), where, in the expanded form on the right side, all flows except the discharging energy are unobservable.

$$\begin{cases} E_{sub} = E_{sub-ess} + E_{sub-train} \\ E_{char} = E_{sub-ess} + E_{train-ess} \\ E_{dis} = E_{ess-train} \end{cases} \quad (5)$$

By simultaneously solving the above equations, further derivation can be obtained as Eq. (6).

$$\begin{cases} E_{ess} = E_{char} - E_{dis} = E_{sub-ess} + E_{train-ess} - E_{ess-train} \\ E_{sys} = E_{sub} - E_{ess} = E_{sub-train} + E_{ess-train} - E_{train-ess} \end{cases} \quad (6)$$

This paper defines the agent state based on the observables in actual TPSS. Additionally, the reward shaping will also be based on the energy relationship between the observables. The state is defined as shown in Eq. (7), where s_d and s_{ind} represent direct and indirect states, collectively forming the state space of the agent. The direct state represents the energy changes of the aforementioned observable variables. The indirect state is used to enhance the agent perception of the environment. It can select information from various dimensions of observable entities based

on actual operation conditions and the capabilities of the agent executing mechanisms, such as the voltage and current of each substation, and the power of each ESS. N and M represent the number of observables in the TPSS, i.e., the number of substations and the ESSs.

$$s \supseteq \begin{cases} s_d = [\Delta E_{sub}, \Delta E_{ess}, \Delta E_{sys}, SOC] \\ s_{ind} = [i_{sub,1}, u_{sub,1}, \dots, i_{sub,N}, u_{sub,N}, P_{ess,1}, \dots, P_{ess,M}, \dots] \end{cases} \quad (7)$$

In the direct state s_d , ΔE_{sub} is defined as the energy consumption of all substations within the selected power supply section during the optimization step, as expressed

$$\Delta E_{sub} = \sum_{i=1}^N \int_0^T i_{sub,i} u_{sub,i} \quad (8)$$

where $i_{sub,i}$ and $u_{sub,i}$ represent the current and voltage of substation i , respectively. T is optimization step, which is the time step for agent action.

ΔE_{ess} is the overall energy change of all ESSs within the optimization step, as shown in the following

$$\Delta E_{ess} = \sum_{k=1}^M \int_0^T P_{ess,k} \quad (9)$$

where $P_{ess,k}$ is the power of ESS k , and charging is positive, while discharging is negative.

ΔE_{sys} is the system energy consumption within the time step, can be obtained from Eq. (10) as following

$$\Delta E_{sys} = \Delta E_{sub} - \Delta E_{ess} \quad (10)$$

Based on the energy relationship derived from Eq. (6), the smaller the system energy consumption E_{sys} , the theoretically greater the regenerative braking energy $E_{train-ess}$ that the ESS may absorb. If E_{sys} is negative, it indicates that the ESS has certainly absorbed regenerative braking energy. Moreover, the above energy relationship is derived based on the principle of energy conservation. Therefore, any changes in environmental characteristics will not affect the derivation process of this relationship, and it will always satisfy the relationship in Eq. (6) at any time step. Therefore, in the control process of the ESS, by real-time calculating the size, polarity, and change of ΔE_{sys} by Eq. (10), it is possible to roughly estimate the absorption of regenerative braking energy, providing certain objectives and guidance for online optimization.

3.3. Charge and discharge guide rules design

The utilization of regenerative braking energy is qualitatively inferred, but shaping an accurate reward based on this remains challenging. However, fuzzy logic excels in expressing qualitative knowledge and experiences with unclear boundaries, simulating human brain-like rule reasoning. Therefore, this paper will attempt to shift from traditional methods to using fuzzy logic rules based on the energy relationship between observables combined with “expert experience” to guide agent learning. To achieve this, the first step involves designing charging and discharging guidance rules for the ESS.

1) Charging guide design

In the process of optimizing control of the ESS, when the ESS is in a net charging state within the optimization step, i.e., when ΔE_{ess} is greater than or equal to zero in Eq. (9). Based on the relationship in Eq. (6), the absorption status of regenerative braking energy can be determined by the system energy consumption ΔE_{sys} within the time step. At this point, there are a total of five scenarios:

a. When ΔE_{sys} is negative, the ESS has necessarily absorbed regenerative braking energy, which aligns with our expectations for the

control effectiveness of the ESS. Therefore, in this case, the agent should be rewarded accordingly.

$$\begin{cases} \text{if : } \Delta E_{sys} < 0 \\ \text{then : } reward \geq 0 \end{cases} \quad (11)$$

b. When ΔE_{sys} is non-negative and relatively large, combined with Fig. 8 and Eq. (6), it indicates that there is inevitably a demand for traction energy in the system, and the ESS being in a net charging state obviously does not meet our expectations.

It is important to note that, due to the scope of this paper covering multiple substations in the power supply section, there is almost no operation condition within a certain time step where the entire power supply section has neither traction nor braking demand [19]. Therefore:

- c. When ΔE_{sys} is non-negative and relatively small, if there is traction demand in the system, allowing the ESS to be in a net charging state also does not meet the expectations.
- d. When ΔE_{sys} is non-negative and relatively small, and at the same time, ΔE_{ess} is relatively large. If there is regenerative braking energy in the system, based on the analysis in the previous part, it indicates that the ESS has absorbed more energy from the substations, and the effective absorption of regenerative braking energy is small. Otherwise, based on Eq. (6), the sum of $\Delta E_{sub-train}$ and $\Delta E_{ess-train}$ is large, which contradicts the presence of regenerative braking energy.
- e. When ΔE_{sys} is non-negative and relatively small, and at the same time, ΔE_{ess} is relatively small. If there is regenerative braking energy in the system, it indicates that the ESS has also failed to effectively absorb it, and the residual part will be eventually lost in the braking resistors.

Therefore, the above scenarios from **b** to **e** do not meet our optimization expectations, so the agent should be penalized accordingly when ΔE_{sys} is non-negative, as shown in Eq. (12).

$$\begin{cases} \text{if : } \Delta E_{sys} \geq 0 \\ \text{then : } reward < 0 \end{cases} \quad (12)$$

2) Discharging guide design

When the ESS is in a net discharging state within the optimization time step, that is ΔE_{ess} less than zero. During the net discharging process, ΔE_{sys} in Eq. (6) is consistently greater than zero, making it impossible to clearly determine the situation of regenerative braking energy. However, further analysis reveals that during the net discharging process, it is possible to roughly assess the traction energy demand in the TPSS based on the ΔE_{sub} . There are a total of three scenarios:

- a. When ΔE_{sub} is large, it indicates a significant traction demand in the system. At this point, the ESS releases the regenerative braking energy absorbed to achieves energy savings and also relieves the burden on the substation. This aligns with our expectations, and therefore, the agent should be rewarded.

Unlike the clear definition of the positive or negative relationship of ΔE_{sys} in the charging guidance judgment, whether ΔE_{sub} is large or small is a relative concept and depends on the choice of the optimization step length. Therefore, to quantify the specific degree of ΔE_{sub} during the net discharging state, we define the energy consumption ratio as follows

$$\lambda = \frac{|\Delta E_{sub} + \Delta E_{ess}|}{\Delta E_{sub}} \times 100\%, \Delta E_{ess} < 0 \quad (13)$$

The value of the energy consumption ratio threshold can be determined based on the actual line conditions and empirical values. In this

work, a value of 70 % is adopted.

Therefore, based on case **a**, it can be concluded

$$\begin{cases} \text{if : } \lambda \geq 0.7 \\ \text{then : } \text{reward} \geq 0 \end{cases} \quad (14)$$

when ΔE_{sub} is small, there are also two scenarios:

- b. When there is a relatively small traction demand in the TPSS, meaning the discharging energy of the ESS is sufficient to cover it, it may be appropriate to consider weakening the discharging capability of the ESS, allowing it to discharge when there is a higher traction demand.
- c. When there is remain regenerative braking energy in the TPSS, the ESS being in a net discharging state clearly does not align with the expected optimization outcome.

Therefore, based on the analysis of scenarios **b** and **c**, it can be concluded

$$\begin{cases} \text{if : } \lambda < 0.7 \\ \text{then : } \text{reward} < 0 \end{cases} \quad (15)$$

3.4. Reward rules design based on fuzzy logic

Unlike conventional DRL approaches that require explicit and well-defined numerical rewards, TPSS exhibits partially observable and highly dynamic conditions, making it challenging to construct an optimal reward function based solely on numerical feedback. Fuzzy logic provides an effective alternative by incorporating expert knowledge into a structured rule-based framework, allowing the agent to infer energy-saving effectiveness from observable variables. By leveraging the observable energy flow relationships in TPSS, fuzzy logic helps guide the agent to learn effective strategies even when full system information is unavailable.

Based on the charging and discharging guide rules combined with the principle of fuzzy logic, the corresponding fuzzy reward and penalty rules can be designed to guide the agent to optimize the charging and discharging control of the ESS in the whole process. The formulated fuzzy charging guide rules have three inputs and one output. The output is the reward “fed” to the agent. The final fuzzy logic rules for charging guidance are presented in Table 1. Based on the principles of fuzzy logic, linguistic variables “NB”, “NS”, “O”, “PS”, and “PB” are defined, representing “negative big, negative small, zero, positive small, and positive big,” respectively. This distinction helps determine the magnitude and polarity of the output reward. The range of reward values is scaled from ± 1 . The first input variable is ΔE_{sys} , directly determining the overall direction of the output reward. To reduce design complexity, linguistic variables “NB”, “O” and “PB”. This distinction helps determine the positive or negative relationship of ΔE_{sys} , thereby assessing the absorption of regenerative braking energy by the ESS. The second and third input variables are ΔE_{ess} and, similarly, the SOC of the ESS in the direct state, represented by linguistic variables “VS”, “M” and “VB” indicating “very small, medium, and very big” respectively.

Table 1
Fuzzy logic guide rules: charging guide.

Fuzzy variable	ΔE_{sys} SOC	NB	O	PB
$\Delta E_{ess} = VS$	VS	O	NB	NS
	M	O	NS	NS
	VB	PS	NS	NB
$\Delta E_{ess} = M$	VS	O	NS	NS
	M	PS	NS	NB
	VB	PS	NS	NB
$\Delta E_{ess} = VB$	VS	PS	NS	NB
	M	PB	NS	NB
	VB	PB	NB	NB

According to the charging guide rules specified in Table 1, when ΔE_{sys} is “NB”, the system energy consumption is negative, indicating the presence of remain regenerative braking energy in the TPSS available for the ESS to absorb. In this case, the ESS is in a net charging state, aligning with our expectations for control optimization results. Therefore, under this condition, the agent should be rewarded accordingly, and the degree of the reward will be determined based on the relationship between ΔE_{ess} and SOC.

When ΔE_{sys} is “PB,” it indicates a significant traction demand within this time interval. The ESS being in a net charging state is highly unreasonable, and therefore, a penalty is applied under this condition. The specific magnitude of the penalty is determined based on the degree of ΔE_{ess} and SOC.

When ΔE_{sys} is “O,” based on the previous discussion regarding scenarios **b** to **e**, the inference is made according to the size of ΔE_{ess} to deduce the traction demand or braking energy absorption in the TPSS. The agent is then subjected to penalties of varying degrees based on the SOC.

Similarly, the formulated fuzzy logic rules for agent discharging guide are shown in Table 2. The established rules output is still the “feed” of reward or penalty to the agent. The first input variable is the energy consumption ratio λ within the time step, defined linguistic variables “VS”, “M” and “VB” representing “very small”, “medium” and “very big” respectively. This is used to assess the traction energy demand in the TPSS under the net discharging state of the ESS. The second and third input variables are the same as the charging guide rules, except that based on the absolute value of ΔE_{ess} to define the size of the net discharging energy of the ESS within the time step.

Finally, the complete guidance for the ESS charging and discharging is implemented, guiding the agent to learn the best energy management optimization strategy. Fig. 9 illustrates the example of whole guide process. The process starts with the observation of key system variables, which are used as inputs for the fuzzy logic-based charging and discharging guides. These guides provide reward signals for the DRL agent, influencing its decision-making process for charging or discharging. And the system transitions accordingly. For example, when the agent optimizes ESS charging control within a time step, if ΔE_{sys} is “NB”, ΔE_{ess} is “VS”, and SOC is “VS”, according to the rules in Table 1, the agent can receive a reward “O”. This indicates that the agent has achieved the goal of absorbing regenerative braking energy after optimization. But, at this point, the charging capacity of the ESS is relatively weak, and ESS still has a large capacity available for regenerative braking energy absorption, the agent should continue to strengthen charging without giving rewards but also penalties.

After formulating the corresponding fuzzy rules, there is also the process of designing and optimizing membership functions (MF) for input and output variables [33]. However, since the discussion on the design of MF is not the focus of this paper, and in general, it can be manually adjusted based on experience, this paper does not delve into the topic due to space limitations. For the input and output variables in Tables 1 and 2, this paper employs continuous MF composed of trapezoidal and triangular functions. The setting of various parameter values is based on empirical manual adjustments without undergoing

Table 2
Fuzzy logic guide rules: discharging guide.

Fuzzy variable	λ SOC	VS	M	VB
$ \Delta E_{ess} = VS$	VS	NB	O	PS
	M	NS	O	O
	VB	NS	O	O
$ \Delta E_{ess} = M$	VS	NB	O	PS
	M	NB	O	PS
	VB	NS	O	O
$ \Delta E_{ess} = VB$	VS	NB	PS	PB
	M	NB	PS	PB
	VB	NB	O	PS

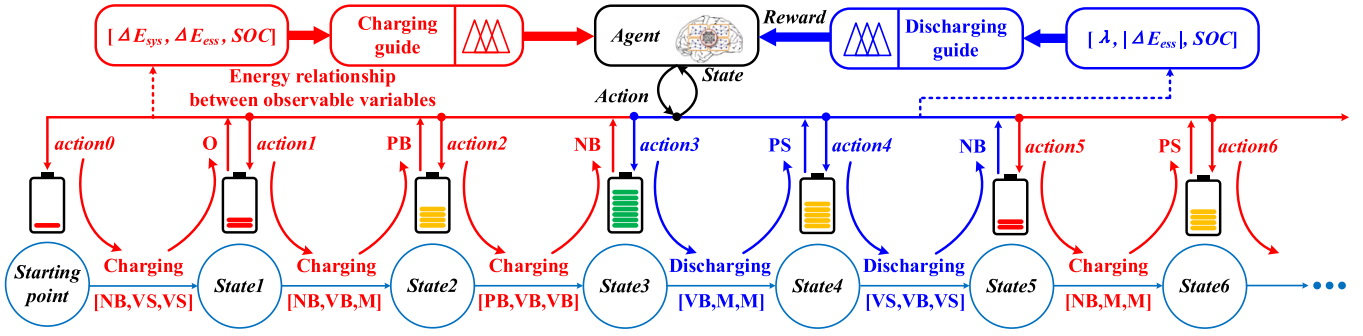


Fig. 9. The guide process of charging and discharging control optimization by agent based on fuzzy logic.

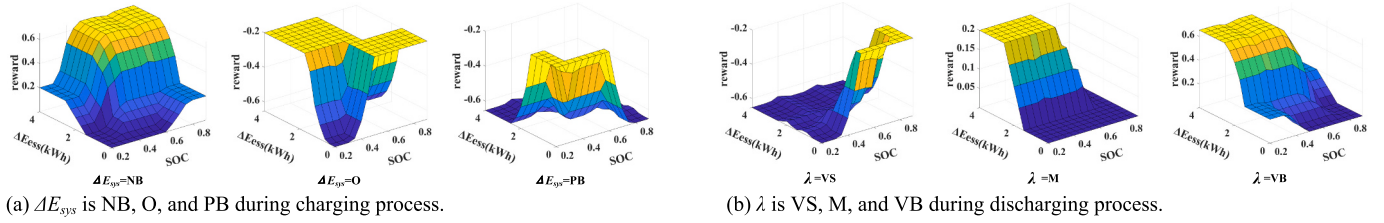


Fig. 10. The input-output relational surface of the fuzzy reward and penalty rules.

additional optimization processes. Fig. 10 illustrates the fuzzy reward-penalty rule surfaces based on the fuzzy logic guided learning method.

The learning process of the agent based on the fuzzy logic guided method is outlined in the following Table 3. The algorithm can undergo pre-training operations based on the simulation environment. The overall process of online learning and deploying is the same as Table 3, with the agent engaging in interaction based on the real-world environment.

4. Case studies

4.1. Case conditions

Based on the actual data provided by the subway operator, the study conducted simulations on a subway line in Beijing. The route of Line is shown in Fig. 11, with a total length of 18.96 km and 13 passenger stations. The simulation model of the TPSS was constructed for the power supply section from the Station V to the Station IX under multiple trains operation condition. The supercapacitor ESS was configured at the Station VII.

The TPSS simulation platform has seen significant development in our previous works [16,19,23]. Due to space limitations, this will not be explained in detail. The conditions of the line, train and ESS parameters are detailed in Tables A1, A2 and A3 in the Appendix, respectively.

4.2. DRL training implementation

The training process of DRL is implemented by the interaction between the agent and the TPSS simulation platform. The agent is constructed by the MATLAB R2021a Deep Reinforcement Learning Box framework. Hardware for training is based on a 64-Core 2.35GHz AMD EPYC 7452 processor with 256GB of RAM. In practice, urban rail trains typically operate with short inter-station travel times, often only a few minutes, and the cycles of traction and braking are even shorter. To balance the rapid energy distribution changes in the TPSS, minimize energy calculation errors, and ensure compatibility with future hardware deployments, the interaction time step between the agent and the environment is set to 10s. This means the maximum number of action steps in one episode is 42.

To validate the applicability of the agent learning method proposed, we tested our method on two different types of DRL methods by the classic DQN and DDPG algorithm. Tables A4 and A5 in the Appendix presents the DRL parameters. Parameters are manually adjusted based on experience, without special design. The ESS adopts a control method based on voltage threshold, as described earlier, controlling charging and discharging by setting the voltage threshold compared with the bus voltage on the traction network side of the ESS. Therefore, the definition of the agent action is the reference value of the voltage threshold for the ESS, as follow

Table 3

The learning process of fuzzy logic guided agent.

Initialize:
1) Initializes the agent online and target parameters;
2) Initializes the experience replay pool;
Execution process:
1) For episode = 1: Maxepisodes
2) Initialize the TPSS state; Initializing agent observation;
3) For time = 1: Ts (<i>Ts</i> is the total simulation time)
4) The action <i>a</i> is output in state <i>s</i> ; Simulation model execution action <i>a</i> (ESS control); Simulation solution gives <i>s</i> _{<i>t</i>+1} at the next time;
5) The energy relationship between observable variables is obtained and input to the fuzzy logic inference module;
6) Agent gets a reward <i>r</i> based on fuzzy guided rules; Store (<i>s</i> , <i>a</i> , <i>r</i> , <i>s</i> _{<i>t</i>+1}) in the experience replay pool;
5) Mini-batch sampling from the Experience replay pool; Online update parameters are calculated based on gradient loss; Update target network parameters based on soft update;
6) End for
7) End for

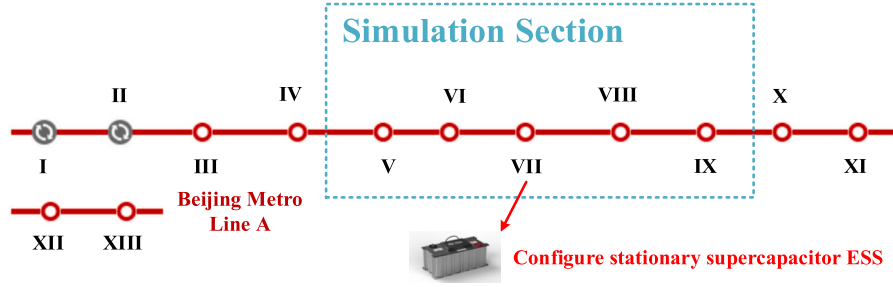


Fig. 11. Diagram of environment configuration.

$$\begin{cases} a = u_{ref} \\ u_{threshold} = u_{ref} \pm \Delta u \\ s.t. \ u_{ref} \in [790, 890] \end{cases} \quad (16)$$

where the hysteresis value Δu between the charging voltage threshold and discharging voltage threshold is set to 5 V. When using the DQN algorithm, the discrete step size of the action u_{ref} is set to 10 V. In the case of by the DDPG, the values of the action are continuous within the range.

To validate the superiority of the proposed method, the EMS based on no-load voltage identification of substation is taken as the baseline strategy. That is, U_{ref} in Eq. (16) takes the no-load voltage of substation, and uses GA to optimize Δu . At the same time, three reward shaping methods were compared respectively based on the DQN and DDPG algorithms:

- 1) The reward is designed based on the energy-saving rate within the sampling step, as per Eq. (1);
- 2) Assuming the train status is known, i.e., the TPSS is completely observable, the reward based on the incremental energy loss of the braking resistor ΔE_{br} within the step;
- 3) The proposed fuzzy logic guide rules.

4.3. Simulation results

A small-scale training consisting of 500 episodes were conducted, with each episode representing a complete train departure interval cycle. Fig. 12 shows the convergence process of braking resistor energy loss by training the DQN agents based on different methods.

Meanwhile, to verify the applicability of the method on different DRL algorithms, we also applied the fuzzy logic guided DDPG for optimizing the ESS control, as shown in Fig. 13. Termination results by different methods are shown in Table 4.

Based on the above results, it is not difficult to find that DRL still represents one of the best methods for the energy management of urban rail ESS. Compared with the baseline strategy, the optimization results

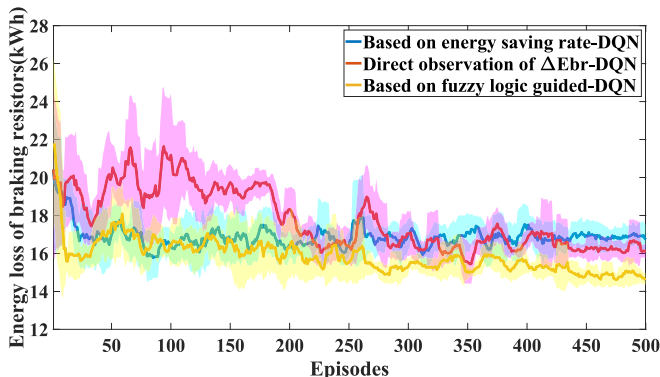


Fig. 12. Optimization results of braking resistor energy loss by DQN agent.

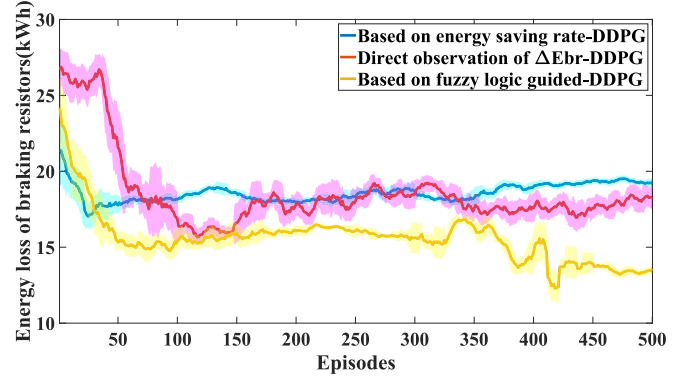


Fig. 13. Optimization results of braking resistor energy loss by DDPG agent.

Table 4

Energy loss of on-board braking resistors.

Method	E_{br}/kWh
Baseline strategy (optimized by GA)	23.4
e-DQN/e-DDPG	17.1/19.5
ΔE_{br} -DQN/ ΔE_{br} -DDPG	15.7/18.1
FLG-DQN/FLG-DDPG	14.2/13.3

based on various DRL methods are significantly improved. In the meantime, the proposed learning method based on fuzzy logic guided can more effectively guide the agent to learn excellent energy management strategy, further reducing the energy loss of the braking resistors. For DQN, the proposed method reduced the energy loss by 39.3 % and 16.9 %, respectively, compared to the baseline strategy and the DQN based on the reward shaping by energy-saving effect. And for DDPG, compared with reward shaping by energy-saving rate and the energy loss of the braking resistor, the proposed method reduced the energy loss by 31.8 % and 26.5 %, respectively.

As shown in Fig. 14, to facilitate the reflection of the utilization effect

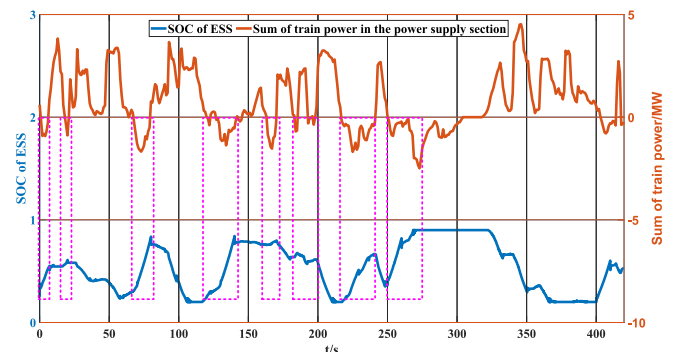


Fig. 14. The effect of regenerative braking energy utilization.

of regenerative braking energy, the sum of the power of all trains in the power supply section is calculated. This simplifies the consideration of the residual power demand after considering the energy interaction between trains. That is, when the power is less than zero, there is regenerative braking energy in the TPSS, and the ESS needs to absorb it; otherwise, it will be consumed in the braking resistors. It can be seen that at this time, the ESS will put into the charging state for absorption as long as it has available capacity. When there is traction demand, the ESS can promptly put into the discharging state, effectively utilizing the absorbed regenerative braking energy, thereby achieving the goal of energy saving.

Fig. 15 (see next page) shows the voltage and output power of each substation when energy management is carried out based on different DRL methods. It can be seen from Fig. (a) that under the control of DDPG agent based on fuzzy logic guided, the parts of traction network voltage exceeding 900 V are significantly reduced. Because of the existence of line resistance, the voltage at the pantograph when the train is in the braking state is generally higher than the voltage of the adjacent substation nodes. Therefore, the starting times of on-board braking resistor are also effectively reduced. In addition, the situation of low voltage of the traction network has also been improved, so the proposed method also has a certain effect in suppressing the voltage fluctuation of the traction network. As can be seen from Fig. (b), the method not only reduces the output power of each substation in the power supply section, effectively reduces the energy consumption of the TPSS. But also reduced the peak power of the substation from more than 5 MW to less than 4 MW. Therefore, compared with the previous methods, the proposed method has better performance in energy saving, voltage regulation and peak power reduction.

5. Conclusions and perspectives

In view of the incomplete observability of TPSS in engineering

conditions and the shortcomings of current evaluation methods of regenerative braking energy recovery, an agent learning method based on fuzzy logic guided is proposed in this paper. Fuzzy logic control is innovatively applied to the agent reward shaping problem, which greatly simplify the reward design process and improve the interpretability in the learning and training process of the agent. The proposed method efficiently learns optimization strategies for ESS only by limited observables in the real environment. The engineering applicability of DRL for energy management of urban rail ESS is further improved. Based on the actual urban rail line conditions, the effectiveness and advancement of the proposed method are verified by case studies. The results show that the proposed method can significantly improve the recovery effect of regenerative braking energy, and thus improve the energy saving performance of the system. For two different types of DRL algorithms, DQN and DDPG, the agent learning method based on fuzzy logic guided can reduce the braking resistance energy loss by 16 % and 31 %, respectively, compared with the traditional methods. In addition, the method is also effective in suppressing the voltage fluctuation of traction network and reducing the peak power of substation. It also shows that this method can be easily extended to the learning process of other different types agents.

Although fuzzy logic provides a practical method to handling partial observability in TPSS, it has certain limitations. First, the fuzzy logic rules are designed based on expert knowledge, which may introduce subjectivity. Second, as the number of fuzzy rules increases, the computational complexity may become a concern, especially when scaling up to larger systems. Finally, the method assumes that the ESS operates independently, and inter-agent collaboration in multi-ESSs is not considered. This could be a direction for future research. In the future, we will explore the performance differences of this method across various DRL algorithms and optimize the fuzzy rule design to further improve the energy-saving performance of urban rail ESS energy management. Additionally, we will continue to explore the application

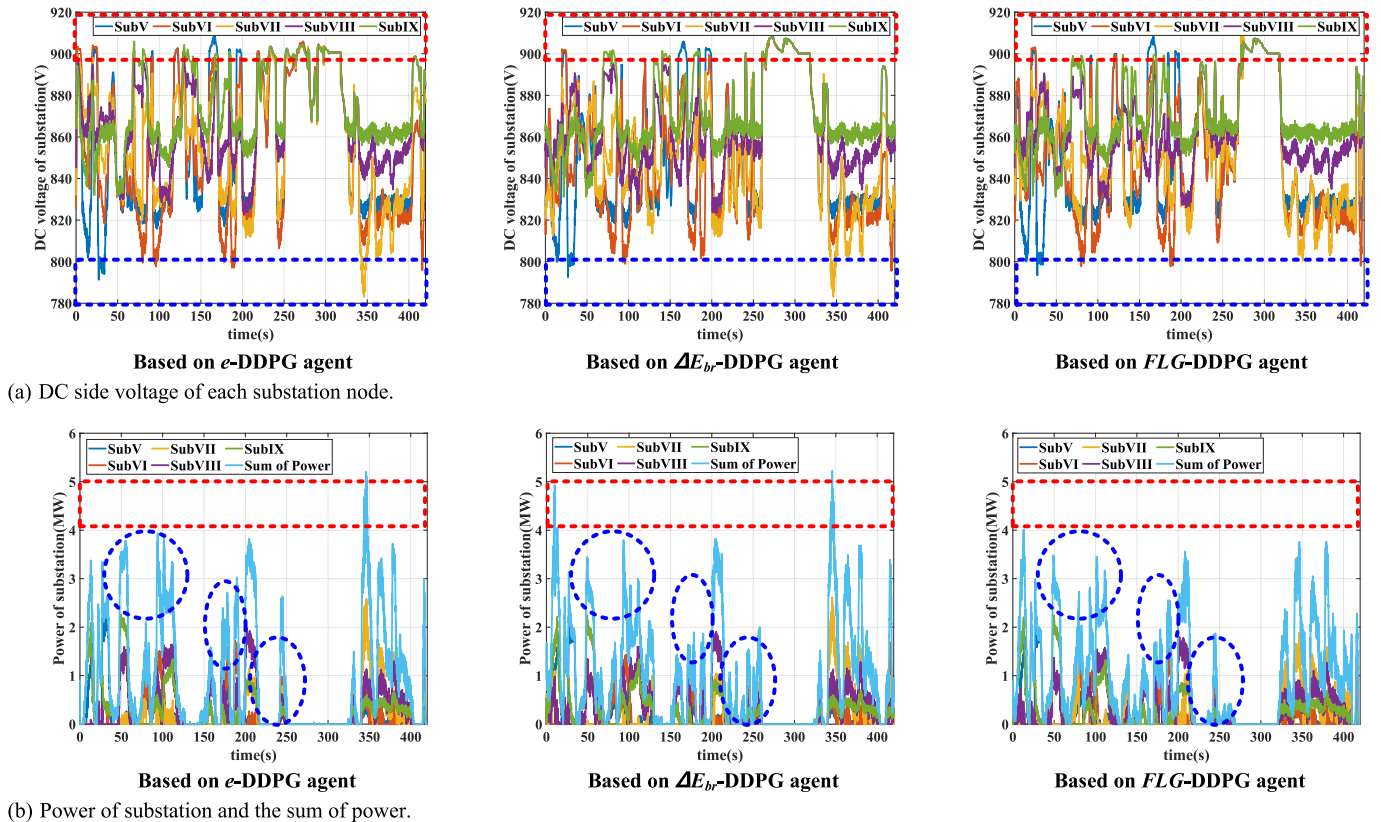


Fig. 15. The voltage and output power of each substation when energy management is carried out based on different DRL methods.

of the proposed method to multi-agent training, where each agent controls a subset of ESSs and collaborates with others to optimize energy management across the entire TPSS. This approach will further enhance the scalability and adaptability of the method in complex urban rail transit environments.

CRedit authorship contribution statement

Yan Li: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Fei Lin:** Writing – review & editing, Supervision, Resources, Investigation. **Zhongping Yang:** Writing – review & editing, Supervision, Funding acquisition. **Xiaochun Fang:** Writing – review & editing. **Xudong Lu:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1

Parameters of line.

Substation	V	VI	VII	VIII	IX
Location (km)	0	1.85	3.54	5.18	6.29
$U_{no-load}$ (V)	829	822	833	859	866

Table A2

Parameters of train.

Parameter	Value
Train formation	3M3T
Weight (AW3)	303 t
Maximum speed	80 km/h
Motor maximum power	265 kW (*4*3)
Motor efficiency	0.92
Starting voltage of on-board braking resistor	900 V

Table A3

Parameters of SC.

Parameter	Value
Rated capacitance	167 F
Maximum power	2.1 MW
Operating range of SOC	0.2–0.9

Table A4

Neural network structure.

Parameter type	Value
Number of hidden layers and neurons number of Critic	256 * 128 * 64
Number of hidden layers and neurons number of Actor	128 * 64
Active layer of the hidden layer	ReLU
Output activation layer of Actor	Tanh

Table A5
Learning parameter.

Parameter type	Value
Optimizer	Adam
Sampling step	10 s
Learning rate	$\theta^Q = 10^{-3}$, $\theta^\mu = 5 \cdot 10^{-4}$
Target network soft update rate	$\tau = 10^{-3}$
Discount factor	$\gamma = 0.99$
Mini-batch size	32
ERP size	10^6
Standard deviation and attenuation rate of noise	$0.3; 10^{-4}$

Data availability

Data will be made available on request.

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